

Multiple Choice

1. **Answer:** B

Options: A) Ridge; B) Lasso; C) both (a) and (b); D) neither (a) nor (b)

Note: Correct option: B - Lasso

2. **Answer:** B

Options: A) $P(A|B)$ decreases; B) $P(B|A)$ decreases; C) $P(B)$ decreases; D) All of above

Note: Correct option: B - $P(B|A)$ decreases

3. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: A - True, True

4. **Answer:** A

Options: A) $O(D)$; B) $O(N)$; C) $O(ND)$; D) $O(ND^2)$

Note: Correct option: A - $O(D)$

5. **Answer:** D

Options: A) Partitioning based clustering; B) K-means clustering; C) Grid based clustering; D) All of the above

Note: Correct option: D - All of the above

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$ '.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'they have high variance'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'PCA'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $p(y|x, w)$ '.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, False'.

Long Form Response

11. **Answer:** def empty_dit(list 1): | return empty_dit

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def check__monthnum__number(monthnum 1): | return True | return False

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key